“Layer su layer”: Identifying and Disambiguating the Italian NPN Construction in BERT's family

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EXPERIMENTAL LAB

MAIN CONTRIBUTION: We investigate whether contextual embeddings in BERT encode constructional (form-meaning) information, using the Italian NPN Cxn as a test case. Through probing experiments, we show that models capture both structural and semantic properties, supporting a constructionist view of language representation.

CxG and PLMs

Interpretability research aims to understand whether the **internal representations** of PLMs encode linguistically meaningful structure.

Cxns are ideal test cases for interpretability [1], as they encode **form–meaning pairings** at the **syntax–semantics interface** [2].

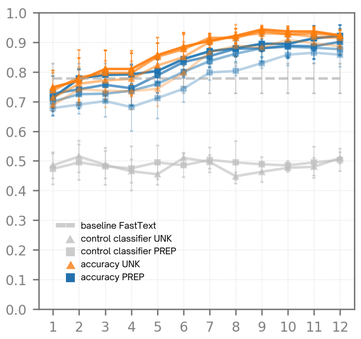
As exposed in a previous study on English [3], **NPN Cxn** is particularly informative: it combines formal constraints with multiple competing meanings and forms a network of **semi-schematic patterns** [4].

This makes it a challenging setting to test whether **PLMs capture structured constructional information**, beyond surface-level lexical cues.

Formal constraints (reduplication, absence of determiners)
Cxn-specific meanings (e.g., succession, contact, accumulation)

The **dataset** [5] is derived from the Italian NPN resource [6] and extended with full sentential contexts from CORIS.

Can BERT distinguish NPN Cxn from distractors?



High accuracy in middle-late layers

UNK ≈ PREP comparable performance with and without lexical cues

Models capture **construction-level structure**, not just lexical cues

FORM OF THE CXN

Identification as a binary classification task

- Positive: NPN Cxns
- Negative: distractors

This tests whether models capture **constructional status**

Data and methods

Distractors counterpart of the dataset [5]:

- **surface-isomorphic pattern**
- **structurally distinct Cxns**.

This allows the task to move beyond purely syntactic discrimination and test whether models capture true constructional status, i.e., form–meaning pairings rather than superficial patterns.

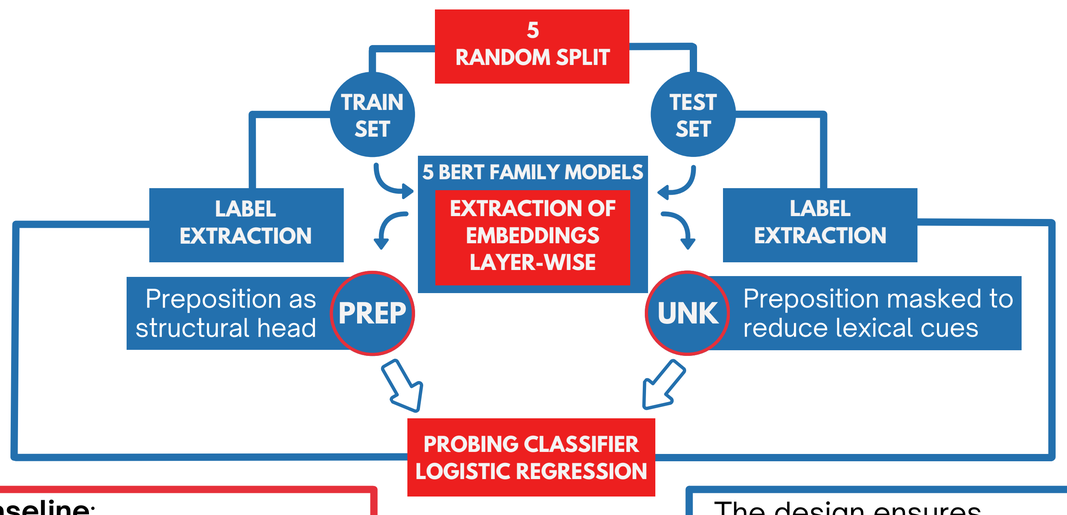
For this study, the analysis is restricted to NPN Cxn (1-4) with the prepositions *a* and *su*

	Form	Meaning	Types	Tokens
1	$[[x]_{N_i}$ *a* $[x]_{N_i}]_{\text{MOD}_j}$	Succession, iteration, distributivity	21	399
2	$[[x]_{N_i}$ *a* $[x]_{N_i}]_{\text{MOD}_j}$	Juxtaposition, contact	27	1108
3	$[[x]_{N_i^{sg}}$ *su* $[x]_{N_i^{sg}}]_{\text{MOD}_j}$	Succession, iteration, distributivity	71	178
4	$[[x]_{N_i^{pl}}$ *su* $[x]_{N_i^{pl}}]_{\text{MOD}_j}$	Greater plurality, accumulation	252	403
5	$[[x]_{N_i}$ *per* $[x]_{N_i}]_{\text{MOD}_j}$	Succession, iteration, distributivity	385	3324
6	$[[x]_{N_i}$ *per* $[x]_{N_i}]_{\text{SUB.CL}_j}$	Inescapable presupposition	21	21
7	$[[x]_{N_i}$ *dopo* $[x]_{N_i}]_{\text{MOD}_j}$	Succession, iteration, distributivity	368	2387
8	$[[x]_{N_i}$ *contro* $[x]_{N_i}]_{\text{MOD}_j}$	Juxtaposition, contact	56	170

DISTRACTORS 989 + CONSTRUCTIONS 1281 → FULL DATASET

Example:
(...)tre donne che scrivono per **donne su donne**(...)
‘(...)three women who write for **women on women**(...)’

Example:
(...)*si battono* ***testa a testa*** *in un torneo*(...)
‘(...)they battle **head to head** in the round(...)’



Baseline:

- Classifier with **static embeddings** (FastText [8])
- **Control Classifier** [7]

The design ensures performance reflects encoded information, not classifier complexity.

Can BERT distinguish NPN’s construction-specific meanings?

MEANING OF THE CXN

Disambiguation as a multi-class classification task

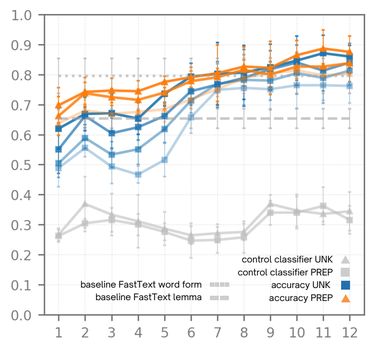
- Succession
- Accumulation
- Contact

This tests whether models capture **constructional meaning**

Performance improves across layers

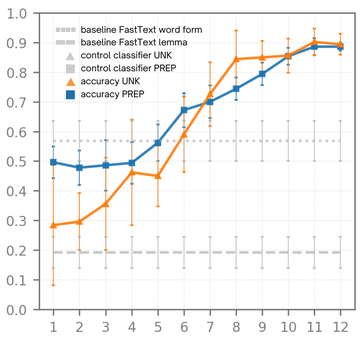
Higher layers encode Cxn’s meaning

Confusions reflect meaning overlap



Models encode **constructional meaning**, but distinctions remain gradient

Semantic generalisation



Strong improvement in higher layers

UNK > PREP in early layers

Models encode **abstract constructional meaning**, not tied to specific prepositions

GENERALISATION TO UNSEEN PREPOSITIONS

TRAIN SET: Distractors + Cxns (1-4)

TEST SET: Cxns (5,7), instanciated by *per* and *dopo*

This tests whether models capture **abstraction** beyond **lexical realisation**

Future directions

⚠ Results remain confined to the representational space within BERT’s models and theoretical implications should be interpreted with caution.

Probing results should be cross-validated with behavioural testing experiments

Empirical ground for theoretical debate on NPN Cxn’s schematisation

Direct comparison between human judgments with model’s predictions

References:

[1] Madabushi, H. T., & Bonial, C. (2025). Construction grammar evidence for how LLMs use context-directed extrapolation to solve tasks. In Proc. 2nd Workshop on Construction Grammars and NLP (CxG+NLP), 190–201.
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[4] Masini, F. (2024). Costruzioni su costruzioni: Idiomaticity and regularity of NPN discontinuous reduplications in Italian. In I fraseologismi schematici, 51–82.
[5] Gorzoni, G., Pannitto, L., & Masini, F. (2026). NPN Construction and Distractor Dataset.
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[7] Hewitt, J., & Liang, P. (2019). Designing and interpreting probes with control tasks. In Proc. EMNLP-IJCNLP 2019, 2733–2743.
[8] Bojanowski, P., Grave, E., Joulin, A., & Mikolov, T. (2017). Enriching word vectors with subword information. In Proc. ACL 2017, 135–146.